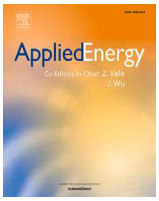
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Large-scale field demonstration of an interoperable and ontology-based energy modeling framework for building digital twins

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HIGHLIGHTS

- Novel ontology-driven framework automates building energy model generation and reduces manual modeling effort.
- Integration between semantic models and energy models enables digital twin adaptability.
- Accuracy and scalability are demonstrated through a hospital case study, achieving mean errors of 0.4°C for temperature, 32 ppm for CO_2, 6% for valve positions, and 4% for damper positions across 45 sensors.
- Automated model generation reduces the need for energy modeling expertise, as demonstrated through rapid scenario testing of building modifications.

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ABSTRACT

Digital twins have emerged as a promising concept for improving building energy efficiency, but their implementation faces challenges in interoperability and adaptability. This paper presents a large-scale field demonstration of an interoperable energy modeling framework for building digital twins, using ontology-based semantic models as data sources for automated model generation and calibration of data-driven component models. The study focuses on a single floor of a hospital building, comprising 12 conditioned zones and data from 45 measuring devices. Across the 45 sensors, the model achieved average mean absolute errors of 0.40°C for temperature, 32 ppm for CO_2 concentration, 0.06 for valve position, and 0.04 for damper position predictions. These results demonstrate the framework's ability to generate and calibrate accurate and flexible building energy models with reduced effort. The paper also showcases the framework's practical application in exploring system modifications to improve indoor comfort, highlighting its potential for scenario analysis and decision support. The proposed approach significantly streamlines the process of creating and maintaining accurate, up-to-date energy models, offering a robust foundation for digital twin applications in the built environment.

1. Introduction

In recent years, digital solutions and the use of Internet of Things (IoT) devices have shown significant potential to improve efficiency and reduce the costs of building operation through advanced services such as performance monitoring and operational optimization. As part of this digitalization, the digital twin (DT) concept has gained traction as a promising approach for managing and utilizing the large amounts of data generated in modern buildings. The DT concept comprises three elements: 1) a physical system, 2) a virtual system, and 3) a bidirectional data flow between the physical and virtual systems [1,2]. Here, the virtual system should act as a digital replica or twin of the physical system, incorporating relevant data and simulation models. These models should be closely integrated, react to changes in the physical twin, and their granularity and accuracy should support the functional outputs or services delivered by the DT [3]. Despite decades of development in building data modeling and simulation, along with significant efforts to improve their interoperability and integration [4–8], they largely remain segregated silos [9,10].

1.1. BIM-based digital twins

Within the built domain, one of the major enabling technologies of the past decades is Building Information Modeling (BIM). The concept

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Nomenclature

$\dot{m}_a$	Mass flow rate of supply and exhaust air to a space
$\dot{m}_w$	Mass flow rate of water supplied to a space heater
$\dot{Q}_{occ,gain}$	Heat generated by people and internal loads correlated with number of people
$\overline{\dot{Q}}_{rad}$	Nominal heat output from a space heater
$\overline{T}_{w,ret}$	Nominal return water temperature of a space heater
$\overline{T}_{w,sup}$	Nominal supply water temperature of a space heater
$\overline{T}_z$	Nominal air temperature at which $\overline{\dot{Q}}_{rad}$ is calculated
Φ_{glob}	Global horizontal solar irradiation
C_a	Thermal heat capacity of air, furniture, and objects contained in a space
C_{bou}	Thermal heat capacity of a generic boundary surface
C_{int}	Thermal heat capacity of interior walls
C_{out}	Outdoor CO_2 concentration
C_w	Thermal heat capacity of exterior walls
f_a	Fraction of solar irradiance absorbed as heat in the indoor air
f_w	Fraction of solar irradiance absorbed as heat in exterior walls
N_{occ}	Number of people in a space
R_{bou}	Thermal resistance of boundary surface between C_{bou} and T_z, and C_{bou} and T_{bou}
R_{int}	Thermal resistance of interior wall between C_{int} and T_z, and C_{int} and $T_{z,adj}$
R_{in}	Thermal resistance of exterior wall between C_w and T_z
R_{out}	Thermal resistance of exterior wall between C_w and T_{out}
$T_{a,sup}$	Supply air temperature
T_{out}	Outdoor temperature
$T_{w,sup}$	Supply water temperature
$T_{z,adj}$	Indoor temperature of an adjacent space

of BIM was initially designed as a file-based data-sharing platform, easing collaboration between various stakeholders during the design and construction phase of buildings [11]. The success and increased adoption of BIM for this purpose have sparked optimism that its capabilities can be extended to the operational phase of buildings, enabling DTs as a central data model. Much of the current research on DTs in the building sector focuses on developing BIM-based solutions and improving the BIM environment to support DTs.

For example, Opoku et al. [12], presented a BIM-based DT platform focusing on real-time indoor climate monitoring of a university living lab case study. The considered variables for monitoring included temperature, humidity, CO_2, and Total Volatile Organic Compounds (TVOC). Based on the historical data collected, static thresholds were considered to detect whether indoor conditions were acceptable.

Peng et al. [13], developed a whole-lifecycle BIM-based DT platform for a large hospital case study. For the operational stage, the study implemented abnormal electricity usage detection using k-means clustering, fault prediction of the air-handling unit using a Long Short-Term Memory (LSTM) model for classification, and a pattern recognition algorithm to analyze the repair messages reported by the technical staff. The study reported 1 % energy savings and a 10 % reduction in facility faults over one year of operation. Moreover, automated actuation and control of the physical system were not implemented due to interoperability issues with the Building Management System (BMS).

Although BIM might prove to play an important role as a valuable data source for DTs, it is currently uncertain whether it can fulfill the role of a central data model. This is because BIM was not originally designed to manage the dynamic and real-time data environments that are characteristic of DTs [2,11].

1.2. Ontology-based digital twins

Recent developments in adjacent IoT-centric domains have highlighted the use of ontologies as a promising concept that can help align and integrate diverse data sources [2,11,14,15]. Ontologies provide a web-based and standardized schema to share, represent, and integrate diverse data sources for improved interoperability [16,17]. The integration of ontologies with BIM models is particularly valuable for applications in the operational phase of buildings. While BIM excels at representing static building information, ontologies provide the semantic framework needed to integrate this information with dynamic operational data. This combination enables more effective digital twin implementations by: 1) Extending the lifetime of BIM models beyond the design phase to building operation. 2) Supporting automated reasoning about building systems and their relationships. 3) Enabling integration with building management systems.

Using BIM as a data source, Merino et al. [18], proposed a data-lake architecture that leverages both the Industry Foundation Classes (IFC) standard and the Brick ontology to integrate data from various sources.

Similarly, Mokhtari et al. [19], developed a cognitive DT for building management and operations, utilizing BIM along with the Building Topology Ontology (BOT) [20] and Brick [21] ontologies. Their work demonstrated the integration of semantic models with various analytics for improved building management, exploring the synergy between BIM and semantic web technologies in DT applications.

Ni et al. [22], demonstrated the use of the RealEstateCore ontology and the Microsoft Azure Digital Twins IoT platform [23] for real-time monitoring of a historic city theater. In a subsequent study, the Brick ontology was further applied to represent systems and equipment [24]. As part of a data analytics layer, different deep learning models were trained and tested for 1-hour-ahead forecasting of heating and electricity consumption.

1.3. Data-driven energy modeling and interoperability

Simulation and prediction capabilities are an integral part of DT solutions and are highly prioritized in other domains with more mature DT development, such as the manufacturing and aerospace industries [2]. However, the building domain is still facing challenges in improving interoperability and reducing implementation costs of building simulation technology [9].

Within the building domain, model types are often categorized as white-box, black-box, or grey-box. White-box models are physics-based, built from domain-specific expert knowledge and first principles. Traditional whole-building simulation programs, such as EnergyPlus [25], fall under the white-box category. On the other hand, black-box models are purely data-driven, relying on machine-learning techniques to extract statistical patterns in data. Grey-box models are a mix of white and black-box models, where physics-based principles are used to define the model structure and data is used to estimate the unknown model parameters.

Current research in the built environment that incorporates predictive features as part of DT solutions primarily relies on monolithic black-box models [13,18,19,24,26], i.e. singular end-to-end black-box models. These models are particularly appealing in the context of DTs because they are straightforward to deploy, require little domain knowledge, and can handle large data volumes effectively. However, their generalization capabilities are often insufficient for tasks such as what-if analysis, retrofitting, and model-predictive control [27,28]. Moreover, the inherent nature of monolithic black-box models offers limited insight and interpretability for decision-makers [29], which is essential for ensuring the model's adoption and trust [30].

In comparison, grey-box models have been shown to generalize better outside the training data region due to the incorporated physics [27,28]. However, grey-box models are currently not as popular for DT applications, as modeling expertise and system topology information is required to define the model structure, making them time-consuming to build and maintain in a DT context. Improving interoperability between data models and simulation models is therefore crucial to overcome these challenges and unlock proven model-based services that can greatly improve building operation efficiency [9,31,32].

1.4. Aim and contribution

This paper addresses the aforementioned challenges by presenting a large-scale field demonstration of an interoperable energy modeling framework for building DTs. The framework is built on previous work [33,34] and combines ontology-based semantic models with data-driven component models to improve the interoperability and adaptability of energy simulation models. It is available as an open-source Python library [35].

Through a comprehensive case study of a hospital building, we demonstrate how this approach can be scaled to complex, real-world scenarios with reduced effort. The contributions of this paper are summarized as follows:

1. **Demonstrating the potential of semantic models:** We showcase how semantic models can serve as effective data sources for the automated generation and calibration of building energy models. This approach significantly reduces both the effort required for modeling and the ongoing maintenance needs, streamlining the process of creating accurate and up-to-date energy models.
2. **Establishing seamless information carryover:** We validate and demonstrate the automated and seamless transfer of information from semantic models to energy models, showcasing their ability to serve as a foundational framework for DT applications. By leveraging semantic models, we enable a streamlined integration that not only improves the fidelity of energy simulations but also supports the dynamic, real-time requirements of DT environments. This approach facilitates the continuous updating and refinement of energy models, aligning them with real-world conditions as they evolve. This capability is essential for the effective deployment of DTs, as it ensures that the virtual representations remain in sync with their physical counterparts.
3. **Proving scalability through a case study:** We validate the scalability of this approach by analyzing a comprehensive case study involving a building with 12 conditioned and thermally coupled zones. The study includes data from a total of 45 measuring devices, against which the model is calibrated. This case study highlights the scalability of the method and its ability to handle complex and diverse real-world conditions.
4. **Applying the generated model in a practical use case:** We apply the generated and calibrated model to a specific use case, demonstrating its practical applications and the benefits of improved interoperability between simulation models. This application highlights how improved model integration can lead to more accurate simulations, better decision-making, and optimized building energy performance.

The paper is structured as follows: In Section 2, an overview of the methodology, scope, and broader context of the paper is provided. In Section 3, the case study is introduced, including relevant information on spaces, equipment, type of control, etc. Section 4 then describes how this information is represented in the developed semantic model of the case study, and how this information is used to generate a simulation model. In Section 5, an automated parameter estimation process is applied to calibrate the generated model and relevant performance metrics are reported. The potential of the method is further demonstrated in Section 6, where the fully adaptable simulation model is used to explore different system configurations to improve indoor climate. Limitations and future work are then discussed in Section 7. Finally, a conclusion with the main findings and reflections is presented in Section 8.

2. Methodology and context

As established, a detailed and accurate energy model is a vital element of DTs to reflect the physical system as closely as possible and unlock services based on predictive capabilities. This paper demonstrates how combining semantic information with data-driven models can produce a fully interoperable and adaptable energy model. This concept is summarized in Fig. 1 within the context of DTs and its three high-level parts: The physical system, the data processing layer, and the virtual system, replicating the behavior of the physical system. For the application shown in this article, two types of information sources are required from the physical system: 1) a semantic model representing the topology of systems, sensors, flow paths, and equipment design information, and 2) operational time-series data from the various measuring devices installed in the real building.

The scheme and structure used to represent the information in the semantic model are specified using an ontology. In this work, the Smart Appliances REFerence (SAREF) ontology [36,37] is used with the extensions SAREF4BLDG [38], SAREF4SYST [39], and the Flow Systems Ontology (FSO) [40]. However, it should be mentioned that the method generalizes to support any ontology.

As shown in Fig. 1, the semantic model should be synchronized with the physical system, reflecting any changes such as equipment replacement, removal, or addition. This allows the simulation model to update accordingly, as is demonstrated later. Based on the information contained in the semantic model, the translator then performs a graph-based pattern match search to identify and assemble the appropriate component models from the component model library to generate a simulation model. Examples of the semantic model and the component model library, related to the case study considered, are provided in Section 4.2 and Section 4.3. After model generation, the model $\mathcal{M}$ is used for simulation, first as part of parameter estimation (bottom), and afterward for inference (top).

A generic user interface layer is also added in the figure to show how a potential user can take advantage of the automated model generation by adding or removing objects to test alternative designs or system configurations. This is explored further in Section 6.

Note that both the user interface layer and data processing layer have been largely simplified in Fig. 1, as they are outside the scope of this work. In addition, integration with other important virtual representations, such as 3D graphical models, is not included. In previous studies, different high-level DT architectures have been proposed to deal with such integrations [18,19,41].

3. Case study

To demonstrate and evaluate the proposed energy modeling and simulation approach, an actual case study building is considered in this paper. The case study is a two-story building within a Danish hospital with a kitchen on the ground floor and office spaces, meeting rooms, and changing rooms on the first floor. The focus of this demonstration is exclusively on the first floor, as illustrated in Fig. 2.

The hospital itself is situated in an open area, while the case study building is next to the main hospital and is exposed to outdoor environmental conditions on most of its outer surfaces.

3.1. Spaces

As shown in Fig. 2, the first floor of the building comprises six office spaces, with floor areas ranging from 18 m^2 to 31 m^2. These offices are equipped with desks, computers, monitors, and multiple lighting sources. The number of occupants varies from single-person offices to shared offices that can accommodate up to six people. Two large meeting

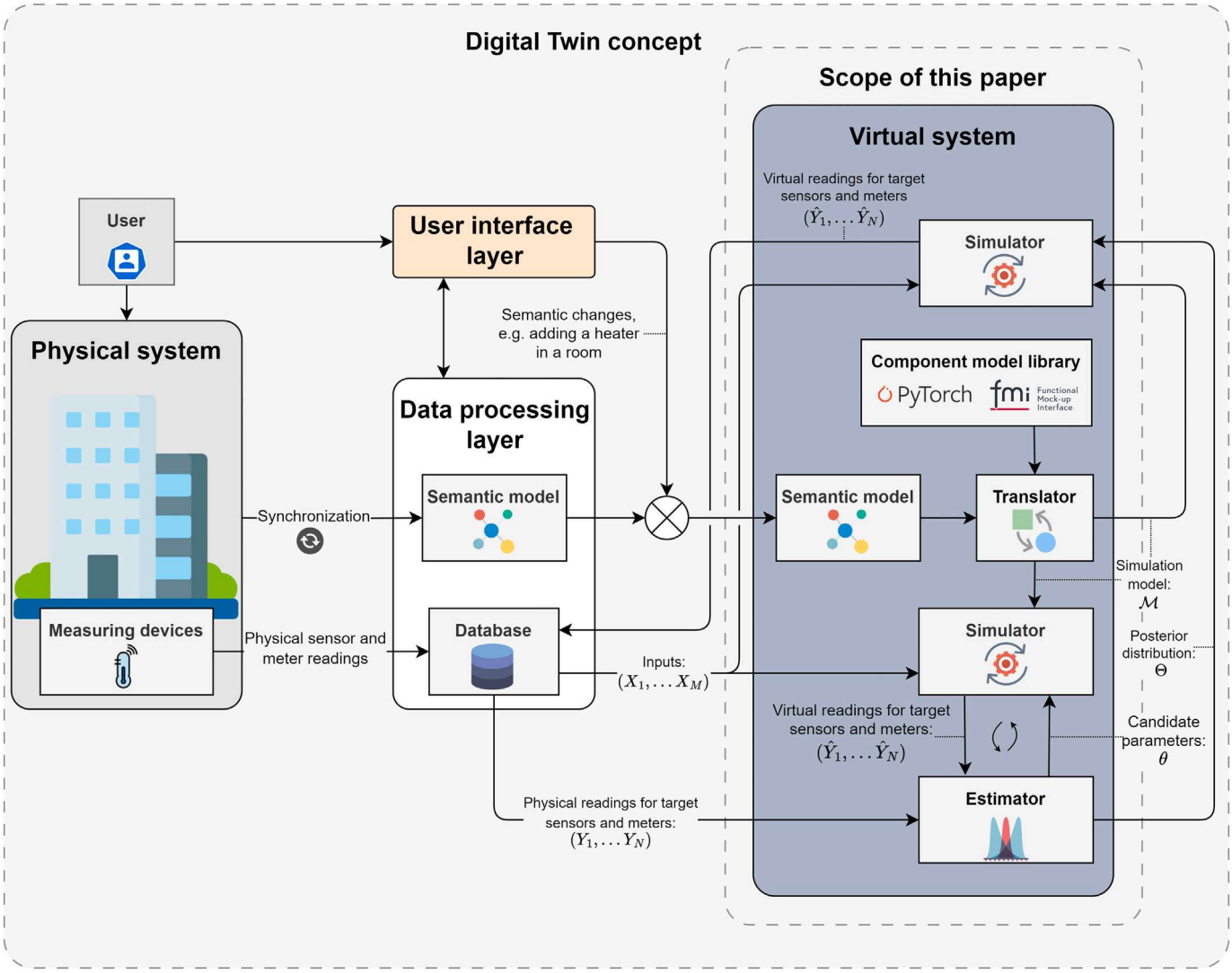


Fig. 1. Methodology overview showing the role of the semantic model and how a simulation model can be continuously adapted to reflect the current state of the physical system or for testing alternative designs or configurations. Adapted from [33].

rooms with an area of 46 m^2, are separated by a folding wall, allowing them to be combined into a single larger space. The first floor also contains two separate changing rooms with areas of 21 m^2 and 42 m^2, these changing rooms have 2 and 6 walled toilet stalls respectively. Furthermore, the floor contains a shared staff accommodation with an area of 82 m^2 and a kitchenette. Other facilities on this floor, such as a stairwell, elevator, and multiple single-stall toilets, are not considered during the demonstration. Table 1 presents a summary of the spaces, including their respective areas, their designated use, and occupancy. Each of the spaces is equipped with multiple sensors and systems, including sensors that measure temperature, CO_2, radiator valve positions, and damper positions. As shown in Table 1, Space 2 does not contain temperature, CO_2, and valve position sensors. These sensor types are installed in all remaining spaces.

3.2. Indoor climate control

As shown in Table 1, demand-controlled ventilation (DCV) is employed in most rooms, regulating air supply flow by actuating the dampers using proportional-integral (PI) controllers to maintain the measured CO_2 concentration within comfortable ranges. All spaces except space 2 are equipped with radiators and motorized valves, also controlled using PI controllers. Most dampers are operated in an on-off fashion to also provide heating in a 0 %–30 % opening position range.

3.3. HVAC systems

The first floor has a dedicated ventilation system with a capacity of 11,000 m^3/h, equipped with a rotating heat exchanger and cooling- and heating water coils. The building is supplied by district heating with an outdoor-compensated supply water temperature, operating between 50 and 60 °C when the outdoor temperature is between −12 and 5 °C.

3.4. Weather data

To provide boundary conditions for the generated simulation model of the case study building, temperature and solar radiation data collected from the Danish Meteorological Institute's weather API [42] are used. The weather data were sampled from one of the nearest weather stations to the hospital.

4. Ontology-driven model generation

This section introduces the semantic model representing the hospital case study, providing concrete examples of its structure and semantics. Following this, an example model, part of the component model library,

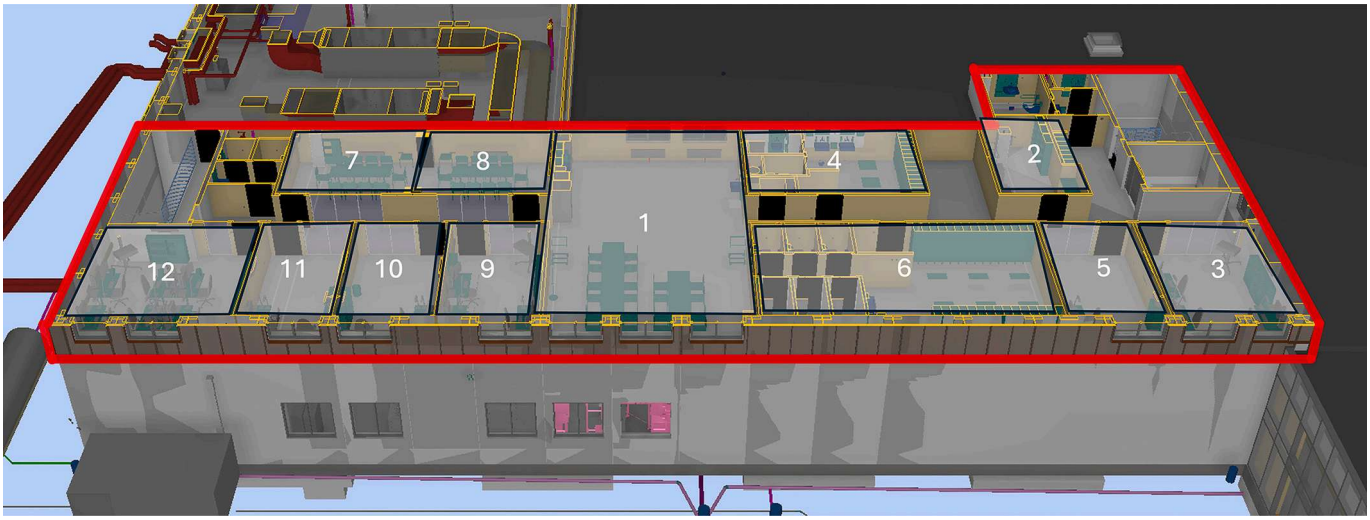


Fig. 2. Graphical overview of the considered floor section of the hospital case study.

Table 1
General information on the spaces included in the demonstration case.

				Measuring devices				Climate control	
Space	Area [m^2]	Type	Occupancy	Temperature	CO_2	Valve position	Damper position	Temperature	DCV
1	82.2	Accommodation		X	X	X	X	X	X
2	15.6	Storage/Copy					X		
3	23.1	Office	3	X	X	X	X	X	X
4	20.7	Showers		X	X	X	X	X	
5	23.1	Office	1	X	X	X	X	X	X
6	41.2	Showers		X	X	X	X	X	
7	46.6	Meeting Room	9	X	X	X	X	X	X
8	46.6	Meeting Room	9	X	X	X	X	X	X
9	23.1	Office	1	X	X	X	X	X	X
10	23.1	Office	1	X	X	X	X	X	X
11	17.5	Office	1	X	X	X	X	X	X
12	30.7	Office	6	X	X	X	X	X	X

is presented along with its semantic representation, denoted as a signature pattern. Finally, using the component models and patterns from the library and the semantic model of the case study, a simulation model is generated and presented.

4.1. Semantic model

A semantic model is employed to represent the physical system in a comprehensive and machine-readable format. The information contained in the semantic model includes the system topology, relationships, and equipment properties of various elements. The semantic model is organized using the so-called triple construct, denoted by `subject` → `predicate` → `object`. Examples of such relations are shown for the case study as a graph in Fig. 3, where subjects and objects are represented by nodes, and predicates are represented by edges. The brackets <> are used to represent an instance of a class, e.g. `<Space_4_space_heater>` as an instance of the class `SpaceHeater`. The figure shows both the full semantic model as well as a highlighted section related to room number 4.

As shown in Fig. 3, the semantic model contains the relevant design data and topological information for the case study described in Section 3. For instance, it contains the following four triples (only triples 2, 3, and 4 are visible in Fig. 3).

1. `<Space_1>` → `connectedTo` → `<Space_4>`
2. `<Space_4_space_heater_valve>` → `suppliesFluidTo` → `<Space_4_space_heater>`
3. `<Space_4_space_heater_valve>` → `isContainedIn` → `<Space_4>`
4. `<Space_4_space_heater>` → `isContainedIn` → `<Space_4>`

The first triple specifies that `<Space_1>` and `<Space_4>` are adjacent while the second triple specifies that `<Space_4_space_heater_valve>` is placed before `<Space_4_space_heater>` in the flow path of the heat carrying fluid. The third and fourth triples specify that `<Space_4_space_heater_valve>` and `<Space_4_space_heater>` are both placed within `<Space_4>`. Using the concept of triples, the whole building can accurately be represented as a large graph or network of objects and relationships. The full semantic model of the case study contains 459 instances of dampers, spaces, controllers, heaters, properties, etc. and 1254 triples connecting these instances.

4.2. Component model library

The component library contains the building blocks from which the simulation model is automatically assembled. It consists of modular self-contained grey-box and black-box models for which inputs and outputs can be connected in different configurations and adapted based on specific use cases. This is achieved using a so-called signature pattern, attached to each component model. The signature pattern acts akin to a blueprint of the component model, describing the semantic context in which the model applies and how it connects to other components.

An example of a component model related to the case study and its accompanying signature pattern is shown in Fig. 4 and Fig. 5. Here,

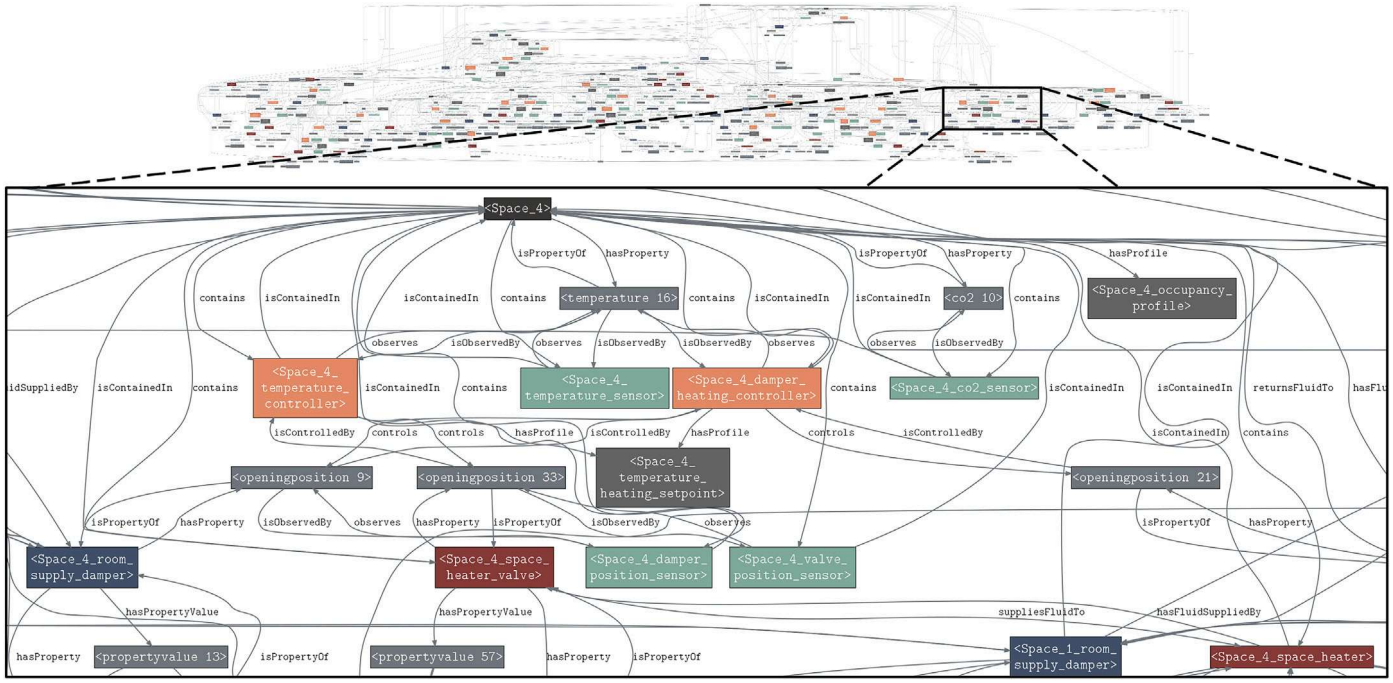


Fig. 3. Semantic model representing the system topology and equipment design information for the hospital case study. The model contains 459 instances and 1254 triples. For readability, a section related to space 4 is enlarged.

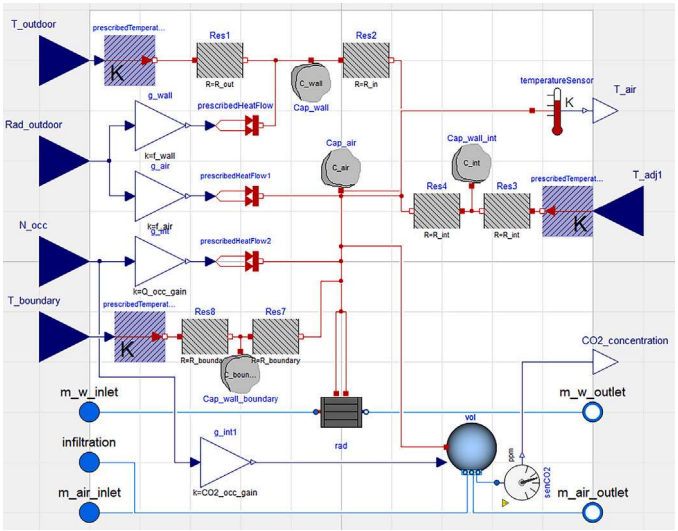


Fig. 4. Example component model of a space.

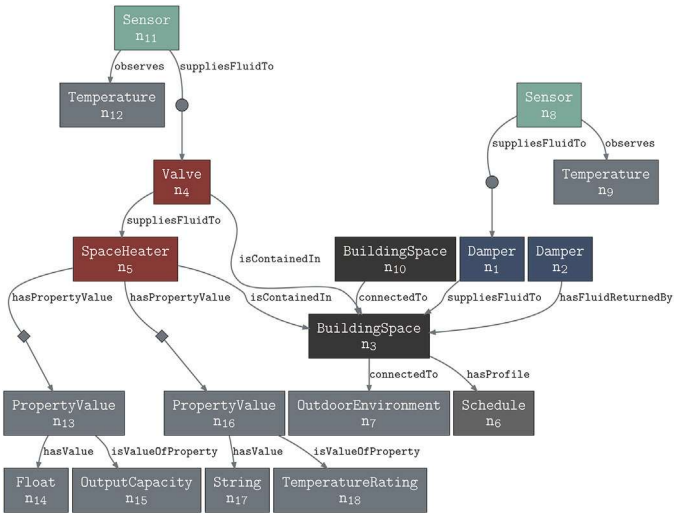


Fig. 5. Example signature pattern specifying the semantic patterns for which the space model applies.

Fig. 4 shows a Modelica diagram representing the heat and mass dynamics of a space, largely based on the grey-box model developed by Blum et al. [32]. While Blum et al. considered a lumped model of a building, we extend this by modeling heat transfer from adjacent zones as well. This is implemented by adding a standard 2R1C pattern for each adjacent zone. Hence, the shown model includes one adjacent space boundary, an outdoor boundary, a space heater, and balanced supply and exhaust ventilation. The accompanying signature pattern shown in Fig. 5 represents the system using the semantics of the applied ontologies SAREF, SAREF4BLDG, SAREF4SYST, and FSO. In this pattern, each node represents one or more ontology classes, e.g. `BuildingSpace`, and is given a local id, e.g. n_3, to distinguish nodes. Below, a subset of four triples, included in the signature pattern, is listed.

1. $\texttt{BuildingSpace}(n_{10}) \rightarrow \texttt{connectedTo} \rightarrow \texttt{BuildingSpace}(n_3)$
2. $\texttt{Valve}(n_4) \rightarrow \texttt{suppliesFluidTo} \rightarrow \texttt{SpaceHeater}(n_5)$
3. $\texttt{Valve}(n_4) \rightarrow \texttt{isContainedIn} \rightarrow \texttt{BuildingSpace}(n_3)$
4. $\texttt{SpaceHeater}(n_5) \rightarrow \texttt{isContainedIn} \rightarrow \texttt{BuildingSpace}(n_3)$

The first triple specifies that the modeled space (n_3), must have an adjacent space (n_{10}). The second, third, and fourth triples specify that a valve (n_4) must be placed before the space heater (n_5) in the flow path of the heat carrying fluid and that they must both be placed in the modeled space (n_3). Comparing these four triples with the four triples of the semantic model, highlighted in Section 4.1, reveals a match. If the remaining triples in the signature pattern can also be matched in

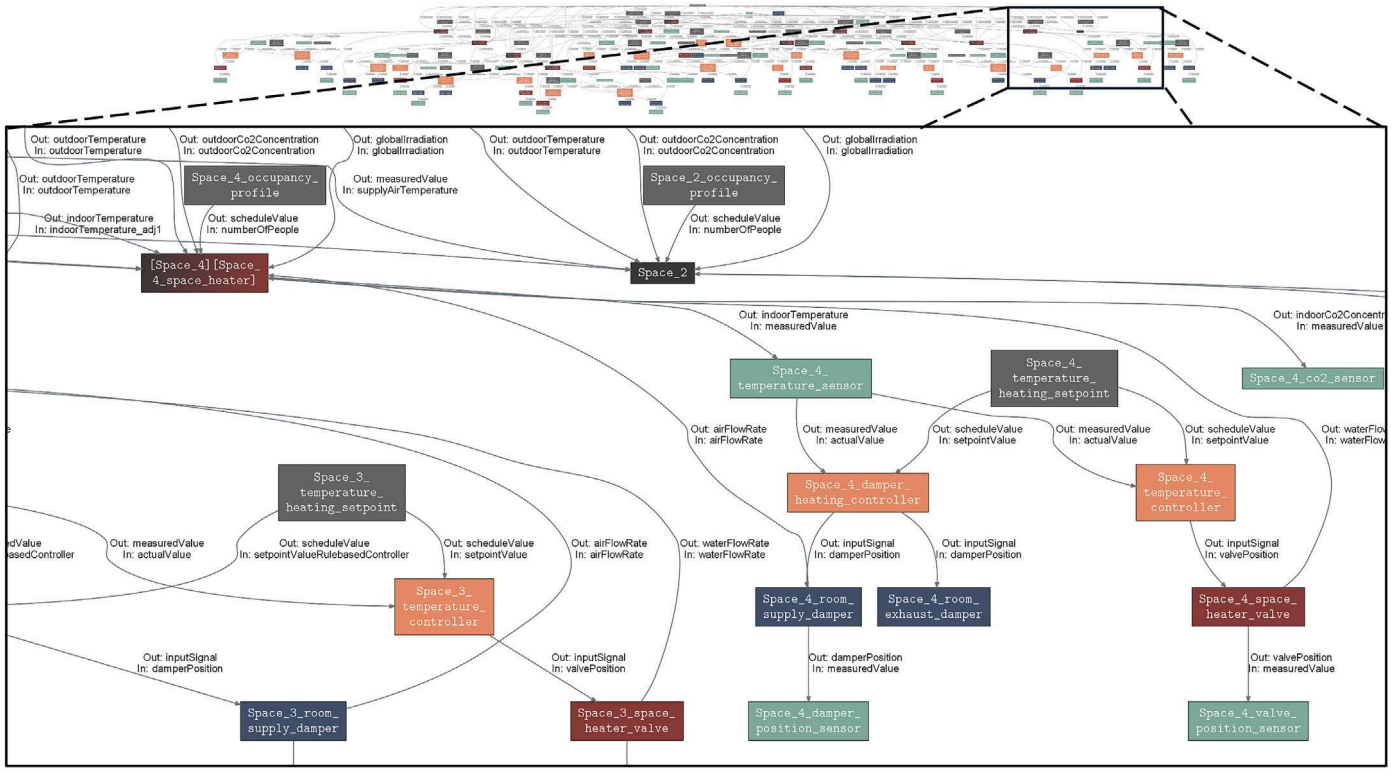


Fig. 6. Simulation model of the hospital case generated automatically from the semantic model using the open source Python package available on GitHub [35]. The model contains 152 interconnected component models of dampers, spaces, controllers, etc. For readability, a section related to space 4 is enlarged.

the semantic model, the component model fits the topology of the physical system. Completing this pattern matching procedure on the various models contained in the component model library, supporting different configurations of spaces, valves, controllers, sensors, etc., produces a set of candidate models.

As the final step, input, output and parameter mapping information is used to assemble the matching component models into one simulation model. For the example space model, this information is provided in Table 2.

As shown in the first column, the model represents the behavior and dynamics of the n_3 and n_5 nodes in Fig. 5. The example model has 9 inputs, including mass flow rates for air and water $\dot{m}_a$ and $\dot{m}_w$, with temperatures $T_{a,sup}$ and $T_{w,sup}$. N_{occ} represents the number of occupants in the zone while the outdoor conditions are provided through the outdoor temperature T_{out}, the outdoor CO_2 concentration C_{out} and the horizontal global irradiation Φ_{glob}. Finally, the temperature of the adjacent space $T_{z,adj}$ is also provided. Each input is mapped from the outputs of other component models as specified in the source column. For instance, the indoor temperature of the adjacent room $T_{z,adj}$ is mapped from the output T_z of the component model representing node n_{10}.

The component model has two main outputs, the indoor temperature T_z and the indoor CO_2 concentration C_z. Model parameters are mapped using information from the last column. Here, C_w, C_{int}, C_{bou}, and C_a are the thermal capacitance of the exterior walls, the interior walls, the floor, and the indoor air. R_{out} and R_{in} are the thermal resistances of the outer and inner sides of the exterior wall while R_{int} and R_{bou} and the thermal resistances of the interior wall and floor. f_w and f_a are the fraction of the solar irradiance Φ_{glob} added to the wall and air, respectively. $\dot{Q}_{ooc,gain}$ is the internal heat gains per occupant. The last five parameters specifies the nominal conditions of the space heater, where $\overline{\dot{Q}}_{rad}$ is the output capacity, $\overline{T}_{w,sup}$ and $\overline{T}_{w,ret}$ is the supply and return water temperature, $\overline{T}_z$ is the indoor air temperature, and n is the heat transfer exponent. As shown in Table 2, $\overline{\dot{Q}}_{rad}$, $\overline{T}_{w,sup}$, $\overline{T}_{w,ret}$, and $\overline{T}_z$ can be mapped from the n_{14} and n_{17} equivalents in the semantic model.

Table 2
Example mapping of inputs, outputs, and parameters of the space model signature pattern.

Nodes	Input		Output	Parameter	
Modeled	Model ⟷	Source		Model ⟷	Source
n_3	$\dot{m}_a$	n_1: $\dot{m}_a$	T_z	C_w	–
n_5	$\dot{m}_w$	n_4: $\dot{m}_w$	C_z	C_{int}	–
	N_{occ}	n_6: y		C_{bou}	–
	T_{out}	n_7: T_{out}		C_a	–
	C_{out}	n_7: C_{out}		R_{out}	–
	Φ_{glob}	n_7: Φ_{glob}		R_{in}	–
	$T_{a,sup}$	n_8: y_{meas}		R_{int}	–
	$T_{w,sup}$	n_{11}: y_{meas}		R_{bou}	–
	$T_{z,adj}$	n_{10}: T_z		f_w	–
				f_a	–
				$\dot{Q}_{ooc,gain}$	–
				$\overline{\dot{Q}}_{rad}$	n_{14}
				$\overline{T}_{w,sup}$	n_{17}
				$\overline{T}_{w,ret}$	n_{17}
				$\overline{T}_z$	n_{17}

4.3. Model generation

The semantic model serves as the basis for automatically generating the simulation model by matching and connecting component models. Performing this procedure on the semantic model of the hospital case study shown in Fig. 3 generates the simulation model shown in Fig. 6. Here, the graph visualizes the information flow and the logical connections between each of the component models in the simulation model. For example,

`[Space_4][Space_4_space_heater]` represents an instance of the example model presented in Section 4.2, showing some of the listed inputs and outputs. For instance, the `[Space_4][Space_4_space_heater]` model receives the input `airFLowRate` (named $\dot{m}_a$ in Table 2) from the `Space_4_room_supply_damper` model.

As shown in Fig. 6, the generated simulation model also contains component models of controllers, dampers, valves, sensors, setpoint schedules, etc. Altogether, the model has 169 unknown parameters. Component models with no inbound edges act as sources or inputs for the simulation model, e.g. the `Space_4_temperature_heating_setpoint` which produces a setpoint signal for the `Space_4_damper_heating_controller` and the `Space_4_temperature_controller`. In addition, measuring devices are directly represented in the simulation model, e.g. `Space_4_temperature_sensor`. These components act as interfaces or outputs of the simulation, enabling a direct comparison with the readings of real sensors and meters.

With the inclusion of controllers in the simulation model, the predictions and estimations in the following sections are all based on a closed-loop configuration. This means that the essential inputs for the model are weather data, setpoint schedules, and occupancy schedules.

5. Bayesian parameter estimation

After model generation, parameters not mapped from the semantic model are estimated using Monte Carlo Markov Chain (MCMC) sampling as described in an earlier work [33]. The goal of MCMC is to sample from the so-called posterior distribution $\pi(\boldsymbol{\theta}|\boldsymbol{Y})$, a distribution of the model parameters $\boldsymbol{\theta}$, given some training data $\boldsymbol{Y}$, as defined in Equation 1.

$$\pi(\boldsymbol{\theta}|\boldsymbol{Y}) = \frac{\mathcal{L}(\boldsymbol{Y}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})}{\int_{\mathbb{R}^{|\boldsymbol{\theta}|}} \mathcal{L}(\boldsymbol{Y}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}} \tag{1}$$

In Equation 1, $\pi_0(\boldsymbol{\theta})$ is the prior density of observing $\boldsymbol{\theta}$, encoding information about the parameters before observing $\boldsymbol{Y}$, typically in the form of domain-specific knowledge. $\mathcal{L}(\boldsymbol{Y}|\boldsymbol{\theta})$ is the likelihood of making observations $\boldsymbol{Y}$, given the parameters $\boldsymbol{\theta}$. This likelihood is calculated by simulating the system with parameters $\boldsymbol{\theta}$ and comparing predictions with actual observations $\boldsymbol{Y}$, assuming certain statistical properties about the system. The integral in the denominator, which evaluates to a constant, ensures that the posterior integrates to 1. This posterior distribution is used to infer posterior predictive distributions, i.e. probabilistic predictions under unseen conditions. The full details are provided in earlier work [33].

The successful convergence of the MCMC sampling process heavily depends on proper parameter initialization, particularly given the high dimensionality of the estimation problem (169 parameters), and geometry and material-specific envelope parameters such as thermal capacitances and resistances. Therefore, the following strategy is used to initialize the parameters of the building envelope, similar to previous work on thermal modeling of buildings [43,44]:

- Thermal capacitances for exterior and interior walls C_w and C_{int} are initialized using wall dimensions and typical material properties
- Air thermal capacitances C_a are calculated from room volumes, also accounting for the thermal mass of furniture [45]
- Thermal resistances for exterior and interior walls R_w and R_{int} are estimated using wall areas, typical material properties, and standard calculation methods [46]

5.1. Parameter sampling

A five-day winter period from December 2023 is used to sample the posterior distribution for the estimated model parameters. The data include measurements from all sensing devices listed in Table 1 along with setpoint logs and weather data. To visualize and summarize the sampled posterior distribution, a corner plot is generated as shown in Fig. 7. For improved readability, Fig. 7 displays a 13 × 13 subset of the full 169 × 169 correlation matrix. This plot includes pairwise correlations between each sampled parameter, marginalized distributions along the matrix diagonal as histograms, and relevant quantile statistics. Pairwise correlation plots can help identify simple correlations between parameters. As shown, for most of the parameters, e.g. infiltration mass flow $\dot{m}_{inf}$ and thermal resistance of interior walls R_{in}, there is no clear correlation. However, for other parameters such as the thermal capacitance of the exterior walls C_w and the thermal capacitance of air C_a, there is a weak inverse correlation, meaning that an increase of C_w can be compensated by a decrease of C_a (and vice versa). The physical interpretation is that both parameters have roughly the same damping effect on the indoor temperature.

5.2. Posterior predictive distribution and inference

To assess the validity of the sampled posterior distribution, the posterior predictive distribution is calculated for the two-week period of the test data. Here, the Average Coverage Error (ACE) is used as a metric to evaluate the reliability of the resulting prediction intervals (PI). The ACE measures reliability by determining the difference between the actual and expected percentages of observations that fall within the specified PI [47]. Hence, if the ACE is negative, it indicates that fewer measurements than expected fall within the given PI range, and vice versa. Additionally, the root mean squared error (RMSE) and mean absolute error (MAE) of the median prediction are also calculated as metrics that can be compared with predictions from deterministic models. Using these metrics, the results are summarized in Table A.4. As shown, the MAE of the temperature point estimates ranges from 0.15°C for space 4 and 0.66°C for space 1. Space 1 is the largest zone with thermal coupling to all other zones on the floor, connected by a long corridor on each side. The more complex thermal dynamics and geometry combined with a higher occupancy variation makes the indoor temperature harder to predict, causing a lower accuracy. The indoor CO_2 predictions have MAE values ranging between 10 and 38 ppm for all spaces except space 12 with a MAE of ∼133 ppm. For some of the rooms, the MAE of valve and damper position predictions are 0 or close to 0, e.g. for space 4. For other rooms, e.g. space 5, the MAE is larger. To show the difference in dynamics causing this, the model predictions are presented for space 4 and 5, over a one-week winter period in Fig. 8(a) and Fig. 8(b). The week shown is part of the test data utilized to evaluate the performance metrics in Table A.4. As shown, the median of the predictive distribution (represented by a black dashed line) aligns with the sensor data (depicted in red) and effectively captures the system dynamics for both rooms and all measuring devices. As shown in Fig. 8(b), the temperature of space 5 fluctuates with the actuation of the dampers u_d and the space heater valve u_v. Most of discrepancies are caused by stochastic occupant behavior, e.g. opening of windows and doors. This causes sudden temperature drops and subsequent activation of the space heater, e.g. as shown Thursday at the end of the period.

On the other hand, there is much less dynamics for space 4. Here, the installed space heater is found to be greatly undersized as the heat delivered is insufficient for maintaining the indoor heating setpoint, which never drops below 21°C during the period shown. This causes the opening position of the space heater valve and dampers to be constantly open. As shown, this is well-captured by the model. For both spaces, the indoor CO_2 concentration is constant at 400 ppm with no significant occupancy.

The PIs of the posterior predictive distribution are also visualized in Fig. 8(a) and 8(b) as a colored area varying from dark grey at 50 % to light brown at 99 %. As shown, the majority of the measurements fall within the 99 % PI. As validation of the PI accuracy, the ACE metric is presented for all measuring devices in Table A.4. For most measuring devices of space 4 and 5, the $ACE_{95\%}$ ranges between −3 % and 5 %, showing the reliability of the estimated PIs. For the valve position PIs of space 5, the $ACE_{95\%}$ is higher at −16 %. This means that 16 % fewer than expected observations fall within the PI.

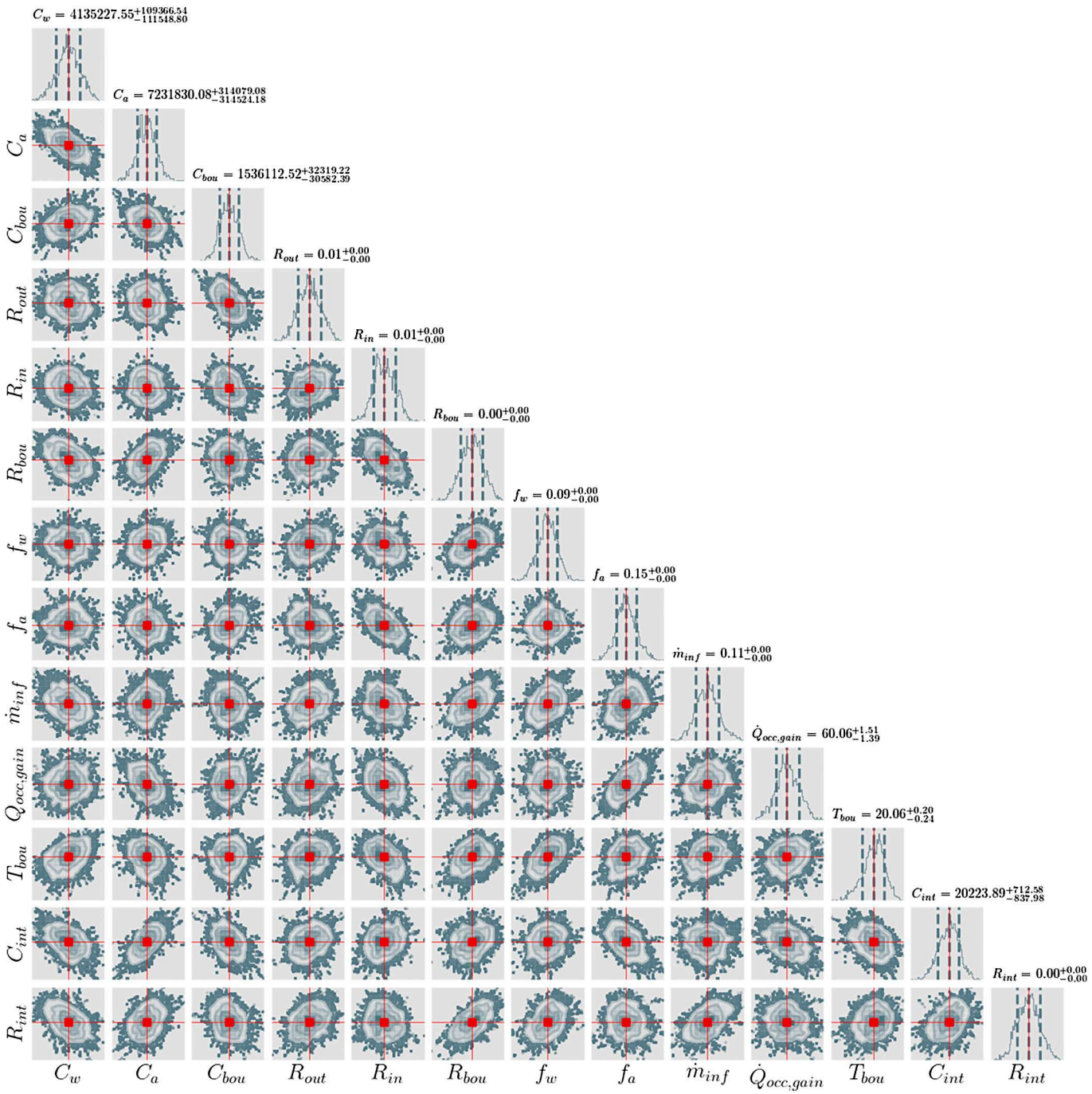


Fig. 7. Corner plot summarizing the posterior distribution of the 13 sampled parameters related to the component model of space 4. The pairwise correlation between the parameters is shown as a matrix plot while the marginalized distribution for each parameter is shown at the diagonal as a histogram. The statistics at the top of the histograms show the quantiles in the format $\theta = Q_{50\%}{}^{+(Q_{84\%}-Q_{50\%})}_{-(Q_{50\%}-Q_{16\%})}$.

6. Adaptive model generation and simulation

This section presents a concrete example demonstrating how changes to the semantic model directly and automatically affect simulation model predictions, illustrating the potential of closely integrating semantic and simulation models. This concept is used to explore alternative system configurations that can improve the indoor environment of the building. As noted earlier, the heater in space 4 is found to be greatly undersized and is constantly operated with a fully open valve. In addition, the room temperature never reaches the desired setpoint of 21 °C (see Fig. 8(a)).

The investigated scenarios are listed in Table 3 where five options are compared, #1: the baseline, i.e. the current system configuration and operation, #2: the current system configuration with an increase in supply water temperature to 65 °C, #3: adding an additional heater of the same model to space 4, #4: adding an additional heater to space 4, but of a larger model, and finally, #5: A combination of #2 and #3 where both a space heater is added and the supply water temperature is increased. For each scenario, the model and specifications of the replaced or added heater is varied according to actual models available on the market [48] and are based on the EN442 standard [49]. The primary

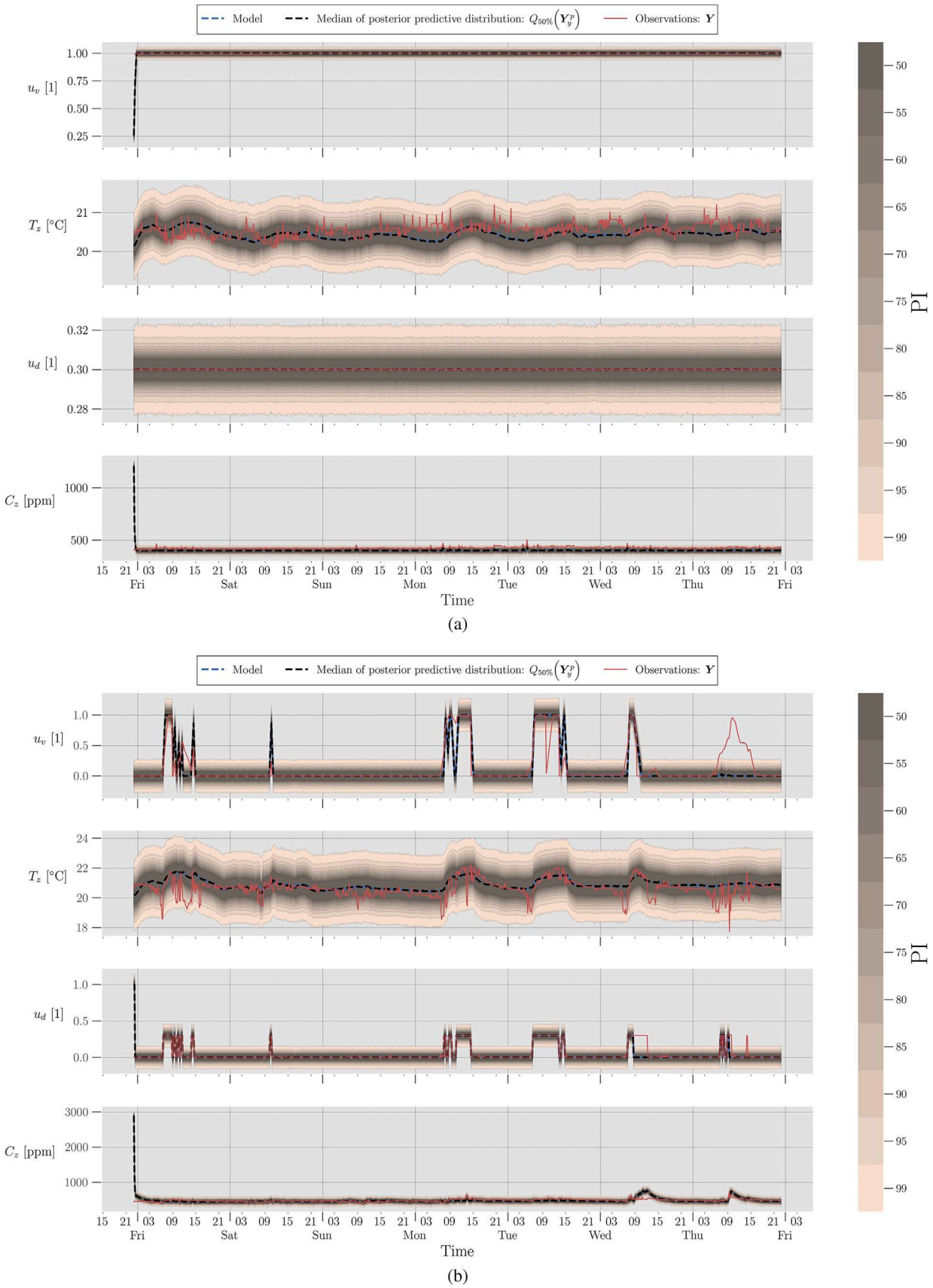


Fig. 8. The median and PIs of the posterior predictive distributions compared with observations for a 1-week period of the test data. **(a)** space 4, a changing room. **(b)** space 5, an office.

Table 3
Scenarios to reduce heating setpoint violations in space 4. The specifications on the radiators reflect actual models available on the market [48] and are based on the EN442 standard [49].

Scenario	Radiator model	Instance	Value
#1: Baseline (Fig. 9a)	1000 × 400 C21	<int 0> <str 0>	432 55/40/20
#2: Baseline, $T_{w,sup}$ = 65°C (Fig. 9a)	1000 × 400 C21	<int 0> <str 0>	432 55/40/20
#3: Add same (Fig. 9b)	1000 × 400 C21 (left) 1000 × 400 C21 (right)	<int 0> <str 0> <int 1> <str 1>	432 55/40/20 432 55/40/20
#4: Add large (Fig. 9b)	1000 × 400 C21 (left) 1000 × 400 C33 (right)	<int 0> <str 0> <int 1> <str 1>	432 55/40/20 747 55/40/20
#5: Add same, $T_{w,sup}$ = 65°C (Fig. 9b)	1000 × 400 C21 (left) 1000 × 400 C21 (right)	<int 0> <str 0> <int 1> <str 1>	432 55/40/20 432 55/40/20

specifications considered are the output heat capacity and temperature classification of a space heater.

The structural differences between the scenarios are shown in Fig. 9, where Fig. 9(a) represents the baseline with one space heater in space 4 and Fig. 9(b) represents scenarios where an additional space heater has been added. Note that the structures are shown in isolation and are still part of the full semantic model of the case study shown earlier in Fig. 3. Besides these structural differences, the information held by the different semantic instances is also varied, as indicated in Table 3. For each scenario, the simulation model is automatically generated based on the data in the semantic model. To evaluate the scenarios, the accumulated deviation from the heating setpoint, given by Equation 2, is used as a comparison metric [32,50,51]. The metric captures both the intensity and duration of setpoint violations and has the unit Kelvin-hours.

$$d = \int_0^T \max(T_{z,set,t} - T_{z,t}, 0)\mathrm{d}t \tag{2}$$

In Equation 2, $T_{z,set,t}$ represents the indoor temperature heating setpoint while $T_{z,t}$ denotes the indoor temperature at time t.

The same two-week testing period discussed in Section 5.2 is also applied here to measure the impact of the suggested scenarios, using the historic heating setpoint signals for all spaces. The results are shown for space 4 in Fig. 10, where Fig. 10(a) shows the calculated discomfort and Fig. 10(b) shows the temperature predictions for each scenario.

As shown in Fig. 10(a), all the alternative scenarios (#2, #3, #4, #5) reduce the indoor setpoint violations and improve the indoor comfort in comparison to scenario #1, the baseline, which has a setpoint violation of 385.62Kh. Scenario #2, with the increased supply water temperature, reduces the setpoint violation to 320.99Kh, a reduction of 17 %. However, as shown in Fig. 10(b), the heating output of the currently installed space heater is still insufficient to meet the desired setpoint. Scenario #3 with an additional heater of 432W, shows a reduction of 74 %, while scenario #4 with a larger space heater of 744W has an even further reduction of 83 %. Finally, scenario #5 shows that an additional space heater of 432W combined with an increase of supply water temperature provides the largest reduction of 85 %.

While the primary aim is to improve the indoor comfort of space 4, the different scenarios impact all considered spaces. For instance, consider the results for Space 6 (Fig. 11), located in the corridor opposite Space 4. As shown in Fig. 11(a), the baseline setpoint violation for Space 6 is 124Kh, which means that the heating capacity currently installed is better sized compared to Space 4. Scenarios #2 and #5, involving

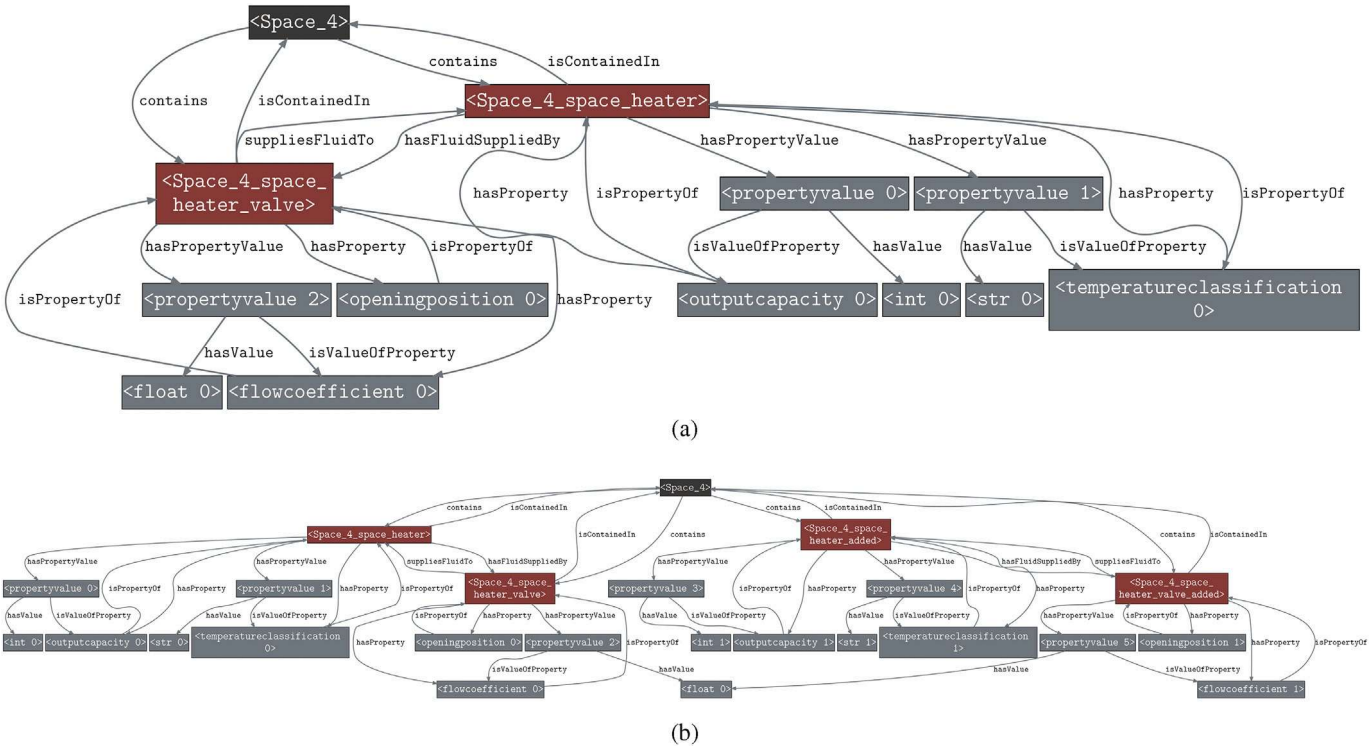


Fig. 9. Structural differences in the semantic model between space 4 with one and two space heaters. Note that the structures are shown in isolation and are still part of the full semantic model of the case study shown earlier in Fig. 3. **(a)** The baseline configuration of space 4 with one space heater (scenarios #1 and #2). **(b)** An alternative configuration of space 4 with two space heaters (scenarios #3, #4, #5).

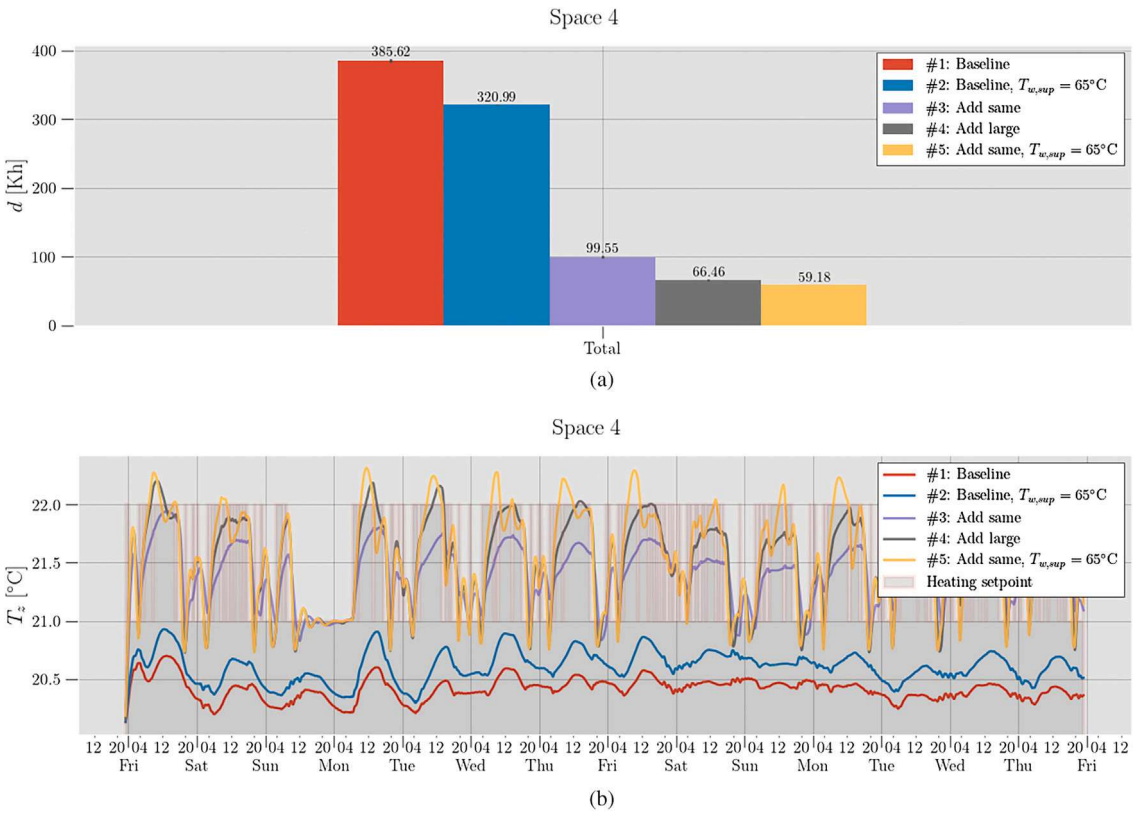


Fig. 10. Impact of four scenarios on indoor comfort in Space 4 over a two-week period. **(a)** Accumulated setpoint violations in Kelvin-hours per scenario. **(b)** Predicted indoor temperatures compared to historical setpoint profile.

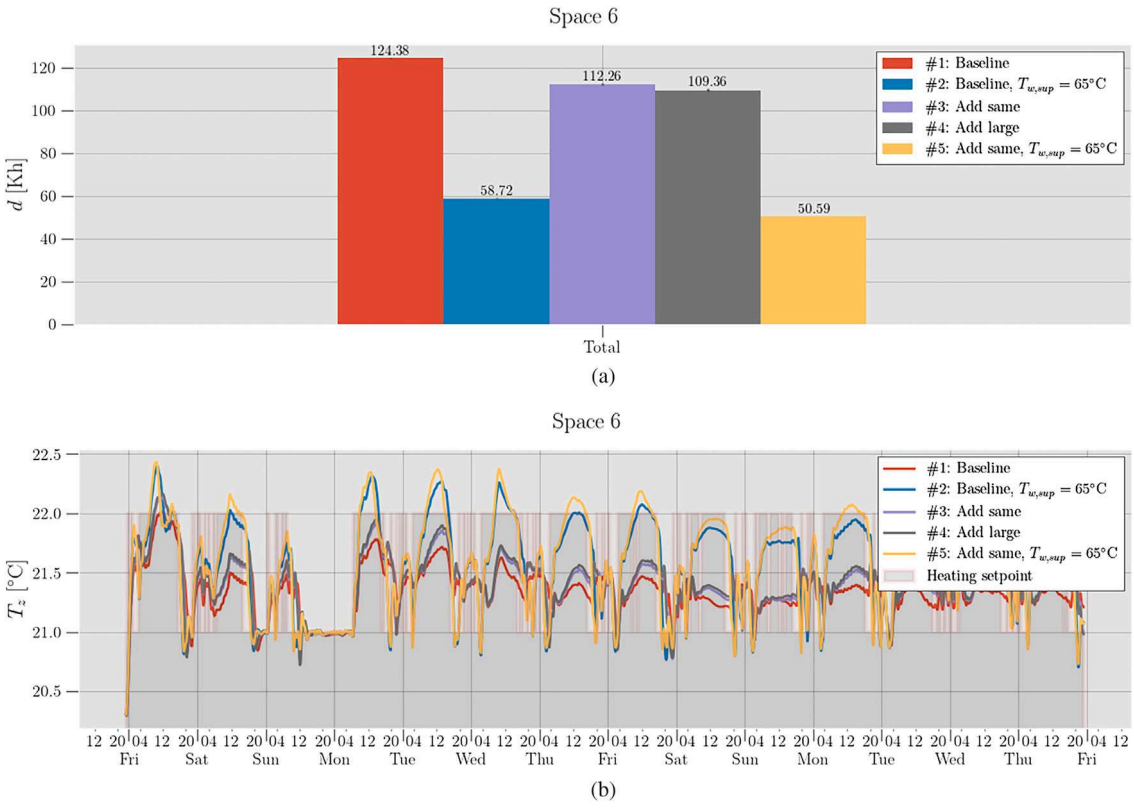


Fig. 11. Impact of four scenarios on indoor comfort in Space 6 over a two-week period. **(a)** Accumulated setpoint violations in Kelvin-hours per scenario. **(b)** Predicted indoor temperatures compared to historical setpoint profile.

increased supply water temperature, have the largest impact, reducing setpoint violations by approximately 50 %.

The generated simulation model also accounts for thermal heat transfer between rooms, revealing secondary effects of the interventions. Scenarios #3 and #4, which focus on adding heaters to Space 4, also slightly improve the indoor comfort in Space 6, reducing setpoint violations by approximately 10 %. This improvement is attributed to the additional heat added in Space 4, creating warmer interior boundary conditions for Space 6. The same effect is also observed for the remaining 10 rooms, although rooms closest to space 4 are affected the most. These observations align with fundamental heat transfer principles, further validating both the generated model and the overall framework.

In summary, significant improvements in the indoor comfort in space 4 can be achieved through simple measures such as installing additional space heater capacity and increasing the supply water temperature. The example presented in this section shows how different system configurations can be explored through changes in the semantic model only. As the semantic model represents the actual physical equipment and system topology of a building, a potential user only needs knowledge of the real system to test strategies. Hence, all abstractions specific to the simulation model (connections, inputs, outputs, parameters, etc.) are hidden away from the user. In comparison, performing such calculations in traditional energy simulation software such as EnergyPlus requires tool-specific expert knowledge, e.g. which parameters or inputs to change and how the different model abstractions map to physical changes in the real building.

7. Limitations and future work

This paper demonstrates the potential of combining semantic modeling with data-driven component models to create a highly adaptable and interoperable energy modeling framework for building digital twins. The approach streamlines the process of generating and calibrating building energy models, and can be scaled to larger cases, as demonstrated through the considered hospital case study. Although the generated model was shown to provide accurate predictions across 45 measuring devices in the case study building, the approach must be further validated on other building types and HVAC system configurations to demonstrate its versatility.

One of the primary automation challenges in this study was the estimation of parameters due to the scale of the case study. Some manual initialization of the MCMC sampler was necessary, particularly for envelope parameters such as thermal capacitances and resistances. Here, high-level geometric data such as areas and volumes were used to calculate initial estimates. To automate this procedure, high-level geometric data and materials could be encoded in the semantic model, e.g., through integration with additional ontologies such as the BOT [20] and potentially BIM models as shown by Mokhtari et al. [19].

Additionally, while this study focused on a winter period, periodic model generation and retraining are likely required to maintain accuracy across different seasons and capture:

Seasonal variations in dynamics and disturbances:

- Summer cooling operations versus winter heating.
- Changed solar gain patterns and their impact on thermal behavior.
- Seasonal differences in occupancy patterns, internal gains, and opening and closing of doors and windows.

Variations in operational regimes:

- Variations in HVAC control strategies across seasons.
- Changes in ventilation requirements between heating and cooling modes.
- System performance degradation over time.

Future work should investigate automated methods for detecting when retraining is needed and determining optimal retraining intervals, possibly building upon earlier work by Blum et al. [32] and Chong et al. [52].

Also, while this paper primarily addresses automated model generation and calibration, it lays the groundwork for broader advancements in building energy management. The automated generation of models presented in this work addresses the time-consuming and costly process of model development, a significant barrier to widespread implementation of model-based services such as MPC [9,31,32].

Based on these observations, promising directions for future research can be summarized as follows:

1. Consider additional case studies with different building types and HVAC system configurations to assess the adaptability of the framework and to further test its limitations.
2. Explore integration with additional ontologies, such as BOT, to access and utilize high-level geometric data for parameter initialization.
3. Investigate continuous generation and calibration of models and how the semantic alignment of the presented framework can support this.
4. Integrating the framework with more sophisticated control strategies and optimization techniques to further improve building energy efficiency and performance.

By addressing these limitations and pursuing these future research directions, the proposed framework can be further improved, potentially leading to more widespread adoption of digital twins and advanced energy management strategies in the built environment.

8. Conclusion

This paper demonstrates the potential of combining semantic modeling with data-driven component models to create a highly adaptable and interoperable energy modeling framework for building digital twins. Through a large-scale field demonstration involving a hospital building, it was shown that this approach can significantly streamline and automate the process of generating and calibrating building energy models while maintaining high accuracy and flexibility. The key contributions of this work include:

1. The successful implementation of a semantic model as a comprehensive data source for automated model generation and calibration, reducing the effort required for modeling and ongoing maintenance.
2. Validation of seamless information transfer from semantic models to energy models, establishing a robust foundation for digital twin applications in the built environment.
3. Demonstration of scalability through a real-world case study involving 12 conditioned zones and 45 measuring devices, proving the approach's viability for complex building systems.
4. Practical application of the generated model to explore system modifications, showcasing the framework's potential for building performance optimization.

The predictions of the calibrated model achieved average mean absolute errors of 0.4 °C for temperature, 32 ppm for CO_2 concentration, 0.06 for valve position, and 0.04 for damper positions across various zones, demonstrating its accuracy and reliability. Moreover, the ability to easily test alternative system configurations by modifying only the semantic model highlights the framework's potential for rapid scenario analysis and decision support. Future work should explore further integration with other ontologies that support high-level geometric data representations as well as the development of advanced services built on top of the generated and calibrated models. Additionally, exploring the application of this framework to a wider range of building types and systems will help to further validate its generalizability and impact.

In conclusion, this research represents a significant step forward in the development of interoperable and adaptable energy models for more widespread adoption of advanced building energy management strategies. This has the potential to improve both building energy efficiency and occupant comfort of buildings.

CRediT authorship contribution statement

Jakob Bjørnskov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Project administration. **August Thomsen**: Software, Investigation, Writing – original draft, Visualization. **Muhyiddine Jradi**: Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Muhyiddine Jradi reports that financial support was provided by Danish Energy Agency. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

Appendix A

Table A.4
Performance metrics of the calibrated model including assessment of the PIs 95 %-75 % for two weeks of test data.

ID	MAE	RMSE	$ACE_{95\%}$	$ACE_{90\%}$	$ACE_{85\%}$	$ACE_{80\%}$	$ACE_{75\%}$
Temperature sensors	[°C]	[°C]	[%]	[%]	[%]	[%]	[%]
1	0.66	0.84	−21	−24	−26	−25	−25
3	0.57	0.7	−5	−8	−11	−11	−12
4	0.15	0.19	4	8	12	14	17
5	0.44	0.63	2	4	8	11	15
6	0.29	0.36	−10	−16	−16	−16	−15
7	0.29	0.4	−7	−6	−5	−4	−3
8	0.44	0.5	0	1	−1	−4	−9
9	0.42	0.49	−1	−2	−1	−2	−4
10	0.3	0.37	4	7	10	12	15
11	0.35	0.44	−1	−3	−4	−5	−4
12	0.45	0.6	−5	−3	−3	−3	−3
Absolute Mean	0.40	0.50	5	7	9	10	11
CO_2 sensors	[ppm]	[ppm]	[%]	[%]	[%]	[%]	[%]
1	10.81	20.34	−2	1	5	8	12
3	22.57	54.8	4	9	13	18	23
4	18.38	27.79	−3	−8	−11	−13	−14
5	26.45	66.5	2	6	11	14	19
6	20.67	23.52	−50	−54	−53	−51	−50
7	38.0	67.5	−6	−4	−2	1	5
8	22.66	42.32	−10	−7	−4	0	4
9	10.02	27.77	5	10	15	20	25
10	33.12	72.36	1	5	8	11	15
11	19.58	41.37	4	8	12	15	18
12	132.56	178.23	−23	−25	−27	−27	−25
Absolute Mean	32	57	10	12	15	16	19
Valve position sensors	[1]	[1]	[%]	[%]	[%]	[%]	[%]
1	0.03	0.07	−50	−45	−40	−35	−30
3	0.12	0.25	−15	−11	−7	−3	1
4	0.0	0.03	5	10	15	20	25
5	0.12	0.26	−16	−11	−7	−2	2
6	0.13	0.21	−10	−12	−11	−9	−6
7	0.01	0.04	0	5	10	15	19
8	0.11	0.15	−1	1	4	7	8
9	0.03	0.1	3	8	12	17	22
10	0.0	0.0	−40	−35	−30	−25	−20
11	0.04	0.11	−1	3	8	12	17
12	0.08	0.18	1	5	9	13	17
Absolute Mean	0.06	0.13	13	13	14	14	15
Damper position sensors	[1]	[1]	[%]	[%]	[%]	[%]	[%]
1	0.0	0.01	5	10	15	20	25
2	0.0	0.02	5	10	15	20	25
3	0.01	0.05	3	8	13	18	23
4	0.0	0.01	5	10	15	20	25
5	0.02	0.07	0	5	10	15	20
6	0.0	0.02	5	10	15	20	25
7	0.09	0.16	0	−10	−10	−6	−1
8	0.1	0.17	−28	−23	−18	−13	−8
9	0.08	0.15	−21	−16	−11	−6	−1
10	0.08	0.15	−6	−10	−6	−2	2
11	0.01	0.04	4	8	13	18	23
12	0.07	0.16	−15	−10	−5	0	5
Absolute Mean	0.04	0.08	8	11	13	14	15

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